

TriMul: contract globally, reuse locally

A channel-wise contraction connects two fused endpoints. Training adds explicit saved tensors and a separate backward pipeline.

A / THE MATHEMATICS

Outgoing

$$T_{ijc} = \sum_k A_{ikc} B_{jkc}$$

Sum over k; one pair matrix for each channel c.

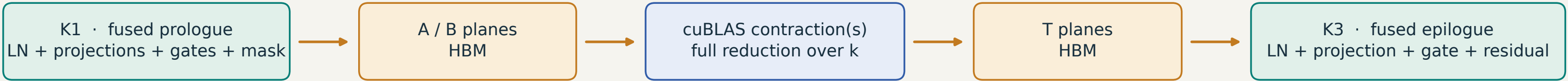
Incoming

$$T_{ijc} = \sum_k A_{kic} B_{kjc}$$

The contracted axis changes; the model semantics stay explicit.

Bidirectional path
separate channel groups
for the two directions

B / THE EXECUTION BOUNDARIES



These HBM intermediates are retained in the packaged inference path. Kernel fusion does not remove the contraction boundary.

INSIDE A K1 / K3 CTA · schematic, not a timing trace

TRANSFER: load 0, load 1, load 2

COMPUTE: compute 0, compute 1, compute 2

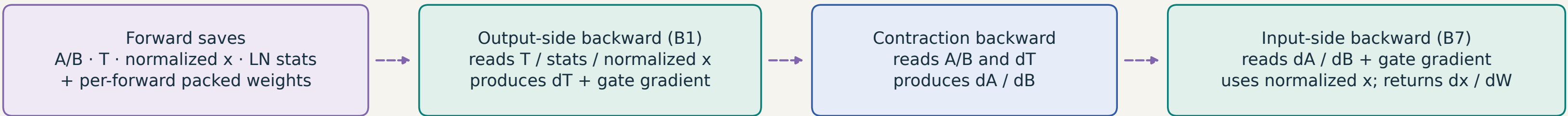
Slot depth and consumer count vary by config.

WHY STOP FUSING HERE?

- Contraction consumes tiles across the k axis.
- CTA-local storage is finite.
- More fusion can raise register / smem pressure.
- A different boundary requires a different algorithm.

The diagram shows current boundaries, not an impossibility proof.

C / TRAINING TENSOR LIFETIMES · D128 PACKAGED PATH



Saved buffers live across forward and backward; gradients and partial reductions may add launches. D256+ research checkpoints are separate.